



C#

Level 1

Assemblies & Metadata (Manifest)



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Definition

Assembly

 An assembly is a compiled .NET unit containing IL, metadata, and possibly resources.

 It is commonly used as a deployment and versioning unit.

Metadata and Manifest

 Metadata describes types and members in the compiled code.

 The manifest describes the assembly itself, including identity, references, and files.

Explanation

Assembly Types

-  `.exe` is commonly used for executable applications.
-  `.dll` is commonly used for reusable libraries.
-  Both can contain IL and metadata.

Assembly Structure

-  IL contains the compiled instructions.
-  Metadata describes types and members.
-  Manifest contains assembly-level information.
-  Resources may also be included.

Procedures

1. Write code using a .NET language.
2. Compile the code into IL and metadata.
3. Store the result in an assembly.
4. CLR loads the assembly.
5. JIT compiles required IL during execution.

💡 Example

MyApplication.exe



References



MyLibrary.dll

➡ Result

The application can use the types and functionality provided by the referenced assembly.

Note

- ◆ Assembly and manifest are not the same thing.
- ◆ Manifest is part of the assembly's metadata.
- ◆ Assembly identity can include name, version, culture, and public key information for strong-named assemblies.

⊘ Warning

- ◆ Do not confuse `.exe` and `.dll` with completely different technologies.
- ◆ Both can be .NET assemblies.
- ◆ Metadata is broader than the manifest.



Comparison

Assembly

- ✓ Complete compiled .NET unit.
- ✓ Contains IL and metadata.
- ✓ Can be an `.exe` or `.dll`.
- ✓ Used for deployment and versioning.

Manifest

- ✓ Part of the assembly metadata.
- ✓ Describes the assembly itself.
- ✓ Contains identity, references, and file information.

✦✦ Summary

- ✓ Assembly = compiled .NET unit.
- ✓ `.exe` commonly represents an executable application.
- ✓ `.dll` commonly represents a reusable library.
- ✓ Metadata describes types and members.
- ✓ Manifest describes the assembly itself.

Thank you 😊